

EduShield - Agentic AI based Student retention prediction system

**PROJECT REPORT FOR THE COURSE
AD23731 FOUNDATIONS OF AGENTIC AI**

Submitted by

SAI SANJAY S V (Reg. No.231501142)
RAVINDERAN K (Reg. No. 231501133)
SUJITH P (Reg. No. 231501163)



**DEPARTMENT OF
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
RAJALAKSHMI ENGINEERING COLLEGE
THANDALAM, CHENNAI, 602105**



RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this is a Bonafide Record of the mini-project work done by **Sai Sanjay S V (231501142)**, **Ravinderan K (231501133)** and **Sujith P (231501163)**, of VII Semester, B.Tech.Artificial Intelligence and Machine Learning Department in partial fulfilment of the requirements of the course **AD23731 – Foundations of Agentic AI** during the Academic Year 2026 – 2027.

Faculty in charge

Submitted for the Practical Examination held on:

INTERNAL EXAMINER

EXTERNAL EXAMINER

TABLE OF CONTENTS

S.No	TOPIC	PAGE No
1.	Introduction	5
2.	Problem Statement and Business Case	7
3.	User and Stakeholder Analysis	9
4.	Proposed Agentic AI Solution	10
5.	Dataset / Knowledge Sources	13
6.	Tools and Technologies Used	14
7.	Agent Architecture	16
8.	Agent Workflow and Orchestration	18
9.	Agentic RAG Implementation	20
10.	System Implementation	22
11.	Experimental Evaluation and Results	23
12.	Comparison, Innovation and Scalability	26
13.	Conclusion and Future Work	29

ABSTRACT

Student retention is a major challenge for educational institutions, where early identification of academic difficulty allows timely support. The proposed project presents EduShield, an Agentic AI-based Student Retention and Academic Intervention System that combines a Multi-Agent architecture, Retrieval-Augmented Generation (RAG), Large Language Model (LLM) reasoning, and Human-in-the-Loop (HITL) decision support.

A Risk Prediction Agent analyzes student attributes such as attendance, marks, and assignment completion, and produces a risk level, a risk score, and the significant risk factors. This output is passed, together with knowledge retrieved from an institutional knowledge base through RAG, to a Retention Intervention Agent, which generates a personalized, prioritized intervention recommendation with action steps and contextual reasoning.

The recommendation is presented to a faculty member in an AI Retention Assistant panel, where it can be approved, modified, or rejected before the final intervention is recorded in the intervention history. EduShield is therefore a decision-support system; the faculty member remains the final decision-maker.

The workflow of the system is: **Student Data → Risk Prediction Agent → Risk Score and Factors → RAG → Retention Intervention Agent → LLM Reasoning → Recommendation → Faculty Review (Approve / Modify / Reject) → Final Intervention → Intervention History**. The previous version of EduShield used an XGBoost model for risk prediction; the upgraded architecture replaces it in the active workflow with the Risk Prediction Agent.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Student retention and academic performance monitoring are important challenges in educational institutions. Students may struggle because of low attendance, poor assessment results, incomplete assignments, or declining academic progress. Faculty members traditionally examine these indicators manually, which is time-consuming and makes personalized support difficult when the number of students is large.

The earlier EduShield system addressed part of this problem through risk monitoring and ML-based prediction. However, a risk label alone does not tell a faculty member why a student is at risk or what action to take. The upgraded system, EduShield v2.0, introduces Agentic AI to move from **risk identification** to **intervention planning**. It uses two specialized agents, a RAG knowledge-retrieval step, LLM reasoning, and HITL review so that faculty members keep control of the final decision.

1.2 Need for Agentic AI

Two students with the same risk level may need different help: one may have an attendance problem, another a marks problem. An intervention must therefore consider the individual student's risk factors and the support guidance that the institution can actually offer.

Agentic AI provides a multi-stage workflow rather than a single prediction or a single LLM call:

Perceive → Analyze → Identify Risk → Retrieve Knowledge → Reason → Recommend → Human Review → Act / Record

Table 1.1: Terminology Used in This Report

Term	Meaning in EduShield
Agentic AI	The overall intelligent workflow from perception to recorded action
Multi-Agent AI	Two specialized agents with separate responsibilities
LLM	The reasoning and generation capability used within the agents
RAG	Retrieval of external institutional knowledge to ground recommendations
HITL	Faculty oversight and final decision

1.3 Project Objectives

The main objective of the project is to develop an **Agentic AI-based student retention system that analyzes academic risk and generates personalized intervention recommendations under faculty supervision.**

The major objectives are:

1. To develop an Agentic AI-based student retention and intervention system.
2. To implement a Risk Prediction Agent that outputs risk level, risk score, and risk factors.
3. To implement a Retention Intervention Agent that generates personalized, prioritized recommendations.
4. To integrate RAG so that recommendations draw on an institutional knowledge base.
5. To use LLM-based reasoning for contextual analysis and recommendation generation.
6. To orchestrate the agents in a sequential, observable pipeline.
7. To implement HITL review in which faculty can approve, modify, or reject recommendations.
8. To store intervention decisions and make intervention history available for monitoring.

CHAPTER 2

PROBLEM STATEMENT AND BUSINESS CASE

2.1 Problem Definition

Institutions hold attendance, marks, assignment, and trend data that indicate which students need help. Turning these indicators into a suitable, student-specific intervention is difficult and effort-intensive. A conventional predictor can flag a student as high risk without explaining the main contributing factors or suggesting an action.

The problem addressed by EduShield is the lack of an intelligent mechanism that can move from student risk identification to context-specific intervention recommendation while maintaining faculty control. The proposed system addresses the following problem:

How can an Agentic AI system analyze student-specific academic information, identify risk factors, retrieve relevant support knowledge, and generate personalized intervention recommendations while faculty retain final decision-making authority?

2.2 Business / Real-World Context

Faculty members and academic coordinators monitor many students at once, often with information spread across several records. EduShield provides one workflow covering student monitoring, risk analysis, knowledge retrieval, recommendation, faculty review, and intervention tracking. It can support academic mentoring, attendance intervention, remedial support, assignment monitoring, study planning, and academic counselling.

2.3 Problem and Proposed Solution

Table 2.1: Problem and Proposed Solution

Problem	Proposed Solution
Large volume of student information	Automated analysis of student data
Difficulty identifying students at risk	Risk Prediction Agent
A risk level alone gives limited context	Risk score and explicit risk factors
Manual, time-consuming intervention planning	Retention Intervention Agent
Difficulty locating relevant support guidance	RAG knowledge retrieval
Generic recommendations	Context-aware, personalized recommendations
Need for human judgement	HITL faculty review
AI output may need correction	Approve / Modify / Reject

The system is therefore designed as a **decision-support solution**, where AI assists the faculty member rather than independently making final academic decisions.

2.4 Business Requirements

Table 2.2: Business Requirements

Requirement	Description
Student monitoring	Monitor attendance, marks, assignments, and related indicators
Risk analysis	Identify academic risk level and score
Risk factors	Identify the main contributing factors
Knowledge retrieval	Retrieve relevant institutional support guidance
Intervention planning	Generate a prioritized recommendation with action steps
Faculty review	Present the recommendation for human review
HITL decision	Allow approval, modification, or rejection
Intervention tracking	Store decisions and provide history

2.5 Expected Benefits

The expected benefits are early identification of at-risk students; recommendations tailored to each student's factors; recommendations grounded in institutional guidance; reduced manual analysis effort; clear separation of agent responsibilities; faculty-controlled decisions; a structured intervention history; and an architecture that can be extended. These are expected benefits only; measured benefits require the evaluation described in Chapter 12.

CHAPTER 3

USER AND STAKEHOLDER ANALYSIS

3.1 Target Users

The primary users of the proposed system are **faculty members and academic coordinators**. Faculty members view student records and risk, trigger the AI analysis, review recommendations, and make the final decision. Counsellor and Student roles are also provided: the login interface offers demonstration accounts for Teacher, Counselor, and Student.

Students are the primary beneficiaries of the system, because it helps faculty identify academic difficulties and provide timely support.

3.2 Stakeholder Identification

Table 3.1: Stakeholders and Responsibilities

Stakeholder	Responsibility
Students	Receive academic support
Faculty members	Review student information and AI recommendations; make the final decision
Counsellors / academic coordinators	Monitor risk and intervention activity
Institution	Support retention and academic performance

3.3 User Requirements

The system must allow a faculty member to: log in securely; see an overview of risk across students; search and filter students by risk level; add, edit, and delete student records; run risk prediction; open the AI Retention Assistant for a student; view the risk level, score, and risk factors; view the recommendation, priority, action steps, and reasoning; approve, modify, or reject the recommendation; and view intervention history.

The recommendation must be presented in a clear and structured form as decision support, not as an automatic academic decision.

CHAPTER 4

PROPOSED AGENTIC AI SOLUTION

4.1 Overall Solution

EduShield v2.0 extends the existing student-retention application with a two-agent pipeline supported by RAG, an LLM, and HITL. The complete workflow is:

Student Data → Risk Prediction Agent → Risk + Risk Factors → RAG → Retention Intervention Agent → LLM Reasoning → Recommendation → HITL → Final Intervention → Intervention History

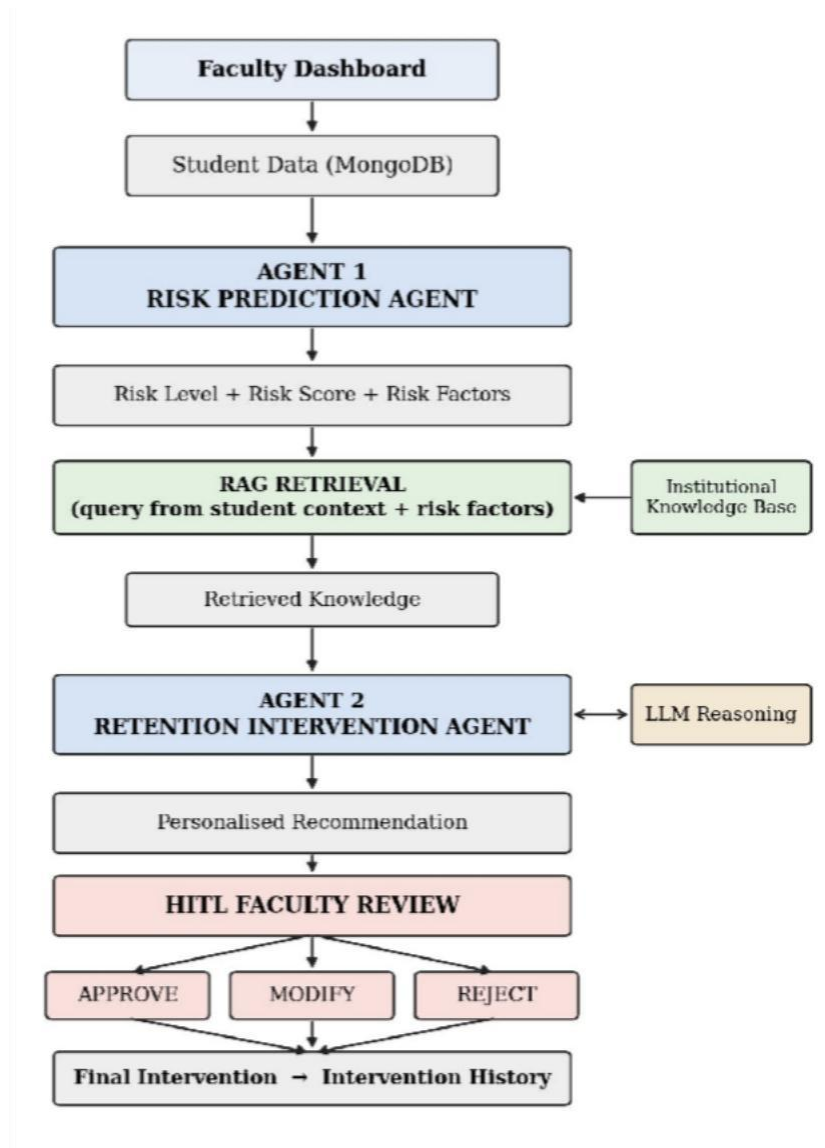


Figure 4.1: EduShield v2.0 Architecture

4.2 Key Features

The major features of the system are:

- Student academic monitoring through a dashboard and student management
- Risk level, risk score, confidence, and risk-factor identification
- Multi-Agent pipeline with visible stage-by-stage status
- RAG retrieval from an institutional knowledge base
- Personalized recommendation with priority, action steps, and contextual reasoning
- LLM-based reasoning
- Faculty review with Approve, Modify, and Reject actions
- Intervention history and on-demand re-analysis

4.3 Multi-Agent Approach

Each agent has one responsibility. The **Risk Prediction Agent** decides how at risk a student is and why. The **Retention Intervention Agent** decides what to do, using the risk information and the retrieved knowledge. Separating the two makes each stage independently inspectable: the interface displays Agent 1 and Agent 2 as distinct sections.

4.4 Risk Prediction Agent

The agent receives the student's available attributes and returns a structured result: risk level, risk score, confidence, and risk factors. In the implemented interface, Agent 1 is labelled *Engine: LLM Reasoning*. For the sample student the result is a HIGH risk level with a risk score of 80% and confidence of 85%, with three identified factors (Section 7.2).

4.5 Retention Intervention Agent

This agent does not predict risk. It receives the student context, the risk level, score, and factors from Agent 1, and the knowledge retrieved by RAG. It outputs a target problem, an AI-recommended intervention, a priority level, a list of action steps, and contextual reasoning that cites the risk output and the retrieved source files.

4.6 RAG Integration

RAG supplies institutional support guidance so that the recommendation is not based on the general knowledge of the LLM alone. Student context and risk factors form a retrieval query; the knowledge base is searched; and the relevant knowledge is passed to the Retention Intervention Agent. The interface confirms the stage "RAG searched institutional knowledge base" and names the source files used in the reasoning.

4.7 LLM Reasoning

The LLM is a reasoning and generation component inside the agents; it is not the whole Agentic AI system. The interface labels the engine of Agent 1 as LLM reasoning, and the output of Agent 2 includes a Contextual Reasoning paragraph that combines the risk

output with retrieved guidance.

4.8 HITL Integration

The recommendation appears under a *Human-in-the-Loop Faculty Review* section with a status badge (PENDING) and three actions: **Approve**, **Modify**, and **Reject**. The panel footer states that the faculty member maintains final decision authority. The system never applies an intervention without this review.

CHAPTER 5

DATASET AND KNOWLEDGE SOURCES

5.1 Student Data

Student records are stored in MongoDB and shown in the Student Management page. The prototype dashboard displays **seven** student records (two High risk, zero Medium, five Low). These records appear to be seeded demonstration data (for example, generated-style email addresses); their origin should be confirmed. Seven records are sufficient to demonstrate the workflow but not to evaluate prediction quality.

5.2 Student Data Categories

Table 5.1: Student Data Categories

Data Category	Purpose
Student profile (ID, name, email)	Identifies the student
Attendance (%)	Identifies attendance concerns
Marks (average, with three component scores)	Represents academic performance
Assignment completion (%)	Indicates task completion; used as a risk factor
Finance category (displayed as BPL / APL)	Socio-economic context shown in the student table
Risk level	Current risk classification
Intervention history	Provides previous support context

5.3 Knowledge Sources

The knowledge base consists of text documents. The source files visible in the output of the implemented system are `academic_support.txt`, `attendance_policy.txt`, `mentoring_guidelines.txt`, `remedial_classes.txt`.

5.4 RAG Knowledge Base

The knowledge base is designed to cover attendance support, academic support, mentoring, remedial learning, intervention procedures, student-support guidance, and progress monitoring. Unless the files are confirmed to be genuine institutional documents, they must be described as **demonstration content** and not as official institutional policy. The recommendation in the sample output refers to an “Attendance Policy”; the corresponding source file was not among those cited (Section 9.7).

CHAPTER 6

TOOLS AND TECHNOLOGIES USED

6.1 Frontend Technologies

A dark-themed web interface built with HTML, CSS, and JavaScript, containing a login page, a Teacher Dashboard with charts, a Student Management table, an Analytics page, and the AI Retention Assistant panel.

6.2 Backend Technologies

Node.js and Express.js provide the API and orchestration layer, as documented in the existing report. Authentication uses JSON Web Tokens (JWT), as stated on the login page, with role-based demonstration accounts.

6.3 Database

MongoDB (MongoDB Atlas per the existing report) stores users, students, and intervention data. The first stage of the AI pipeline reads “Student data retrieved from MongoDB”.

6.4 AI Technologies

Multi-Agent AI, LLM reasoning, RAG, and HITL, described in Sections 6.5 to 6.8. The earlier XGBoost model belongs to the previous architecture.

6.5 Agentic AI

The workflow Perceive → Analyze → Identify Risk → Retrieve → Reason → Recommend → Human Review → Record is implemented as a sequential five-stage pipeline whose progress is shown to the user.

6.6 RAG

Retrieval over text knowledge documents. The retrieval technique used for this project is Keyword matching.

6.7 LLM

Provides reasoning and generation inside the agents.

LLM Provider: Google AI (Google AI Studio / Google Generative AI)

Model: gemini-2.0-flash (Configurable via `process.env.AI_MODEL`)

6.8 HITL

Approve, Modify, and Reject controls with a review-status badge, so that faculty hold final authority.

6.9 REST APIs

The frontend communicates with the Node.js/Express backend through REST APIs. The specific routes for prediction, analysis, decisions, and history are Direct HTTP REST API

Table 6.1: Technology Stack

Technology	Purpose
HTML, CSS, JavaScript	Web interface and interaction
Node.js, Express.js	Backend runtime and API
MongoDB / MongoDB Atlas	Student and intervention data storage
JWT	Authentication and role-based access
REST APIs	Communication between frontend and backend
LLM	Reasoning and recommendation generation
RAG over text knowledge files	Institutional knowledge retrieval
Multi-Agent pipeline	Specialized risk and intervention agents
HITL interface	Faculty approval, modification, rejection
XGBoost (legacy)	Previous risk predictor; baseline only if retained

CHAPTER 7

MULTI-AGENT ARCHITECTURE

7.1 Overall Agent Architecture

The architecture is shown in Figure 4.1. It contains the Faculty Dashboard, the student data layer, the Risk Prediction Agent, the RAG retriever and knowledge base, the Retention Intervention Agent, LLM reasoning, the HITL review interface, and the intervention history.

Table 7.1: Major Component Responsibilities

Component	Responsibility
Faculty Dashboard	Displays students, risk, and the AI Retention Assistant
Student Database	Provides student information; stores decisions
Risk Prediction Agent	Determines risk level, score, and factors
RAG Retriever	Retrieves relevant institutional knowledge
Retention Intervention Agent	Generates the personalized recommendation
LLM	Performs contextual reasoning and generation
HITL Interface	Captures the faculty decision
Intervention History	Records final decisions

7.2 Risk Prediction Agent

For the sample student Arjun Kumar (STU0001), the implemented interface shows a predicted risk level of HIGH; a risk score of 80% (stated as 0.8 in the reasoning text); a confidence of 85%; and the risk factors *Critical attendance deficit (49%)*, *Low academic performance (Test Avg: 54)*, and *Low assignment completion (0%)*. The Student Management table shows attendance of 49% and a marks average of 54 for the same student, consistent with the factors reported.

7.3 Retention Intervention Agent

For the same student the agent reports a priority of *Immediate*; a target problem listing the three risk factors; the recommended intervention *Schedule urgent mentor meeting regarding attendance recovery*; and four action steps (urgent mentor meeting; daily attendance tracking sheet; additional practice worksheets and peer study support; scholarship eligibility review and book bank access). Its contextual reasoning cites the output of the Risk Prediction Agent and retrieved guidance from `academic_support.txt` and `student_support_guidelines.txt`.

7.4 Agent Responsibilities

Agent 1 owns risk classification and the explanation of why a student is at risk. Agent 2 owns intervention selection and the explanation of what to do and how. Neither agent applies a decision; the faculty member does.

7.5 Agent Inputs and Outputs

Table 7.2: Multi-Agent Responsibilities

Agent	Input	Responsibility	Output
Risk Prediction Agent	Student attributes from MongoDB (attendance, marks, assignment completion, related fields)	Assess academic risk and identify contributing factors	Risk level, risk score, confidence, risk factors
Retention Intervention Agent	Student context, risk output, retrieved knowledge	Select and justify a personalized intervention	Target problem, recommended intervention, priority, action steps, contextual reasoning, source references

7.6 Agent Interaction

The agents do not call each other directly in the described design; the backend orchestrates them. The structured output of Agent 1 is combined with the retrieval result and supplied as input to Agent 2 (Chapter 8).

7.7 LLM Integration

LLM reasoning is the engine of Agent 1 as labelled in the interface, and supports the contextual reasoning of Agent 2. Because LLM output can vary between runs and has not been validated against ground-truth outcomes, risk predictions must be evaluated and compared with the legacy baseline before accuracy is claimed (Chapter 12).

7.8 HITL Integration

The output of Agent 2 is routed to the faculty review section rather than to any automatic action (Chapter 10).

CHAPTER 8

AGENT WORKFLOW AND ORCHESTRATION

8.1 Workflow Design

The workflow starts when a faculty member opens the AI Retention Assistant for a student in the Student Management page. The panel then runs a five-stage pipeline and displays each stage as completed:

1. Student data retrieved from MongoDB
2. Risk Prediction Agent evaluated attributes
3. RAG searched institutional knowledge base
4. Retention Intervention Agent generated recommendation
5. Waiting for Faculty Review (Human-in-the-Loop)

The faculty member can re-run the pipeline using the *Re-analyze* action and can open previous decisions through the *History* button, which shows a count (2 for the sample student).

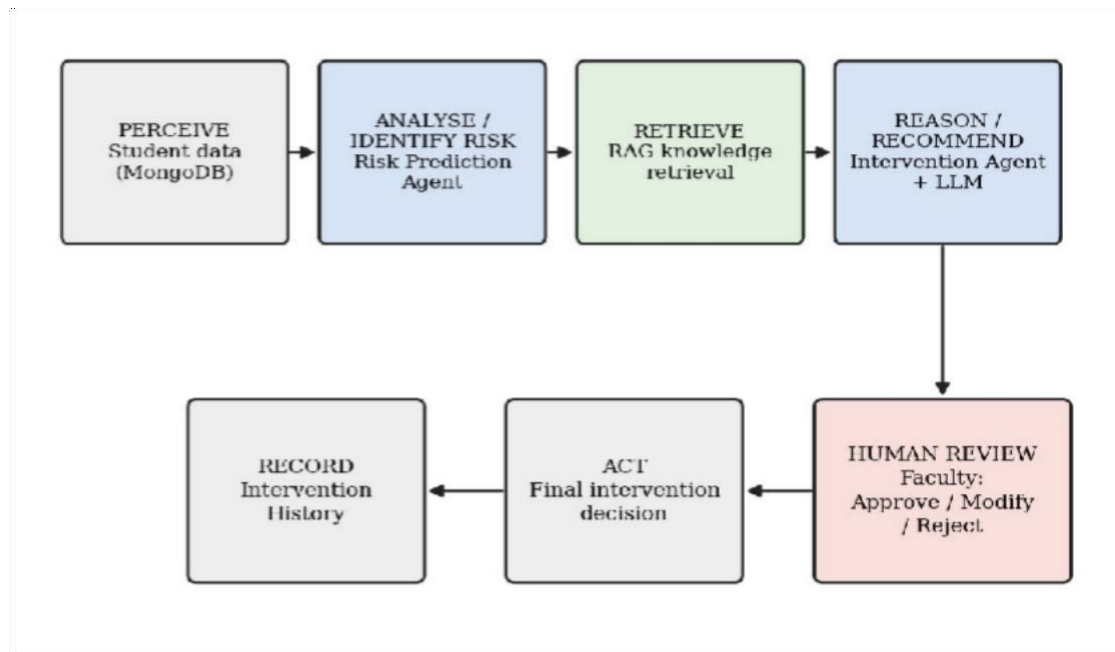


Figure 8.1: Agentic Workflow (Perceive to Record)

8.2 Task Decomposition

Table 8.1: Task Decomposition

Task	Responsible Component
Student data retrieval	Backend / MongoDB
Risk analysis and risk-factor identification	Risk Prediction Agent
Retrieval query formation and knowledge retrieval	RAG retriever
Contextual reasoning and recommendation	Retention Intervention Agent + LLM
Review and final decision	Faculty (HITL)
Decision storage and history	Backend / MongoDB

Decomposing the task allows each stage to be verified separately instead of trusting a single end-to-end model output.

8.3 Agent Coordination

Coordination is sequential and data-driven. Each stage receives the structured output of the previous stage: Agent 1 produces the risk level, score, and factors; RAG converts these into a query and returns knowledge; Agent 2 combines all three with the student context. The two agents therefore share information only through the orchestration layer.

8.4 Orchestration

The Node.js/Express backend orchestrates the sequence, consistent with the specification that the upgrade should extend the existing architecture rather than introduce additional microservices. The user interface shows orchestration progress through the stage list in Section 8.1.

8.5 Decision Flow

- If the student's risk is HIGH, Agent 2 assigns a priority (the sample shows *Immediate*) and produces a multi-step plan.
- If risk is LOW, the specification requires contextual analysis rather than assuming that low risk means no action.
- If retrieval or LLM generation fails, the system must report the failure explicitly rather than claim that knowledge was retrieved or present fabricated output.

8.6 HITL Workflow

After Stage 5 the recommendation awaits review with status PENDING. The faculty decision changes the status and produces the final intervention record (Chapter 10).

CHAPTER 9

RAG IMPLEMENTATION

9.1 RAG Architecture

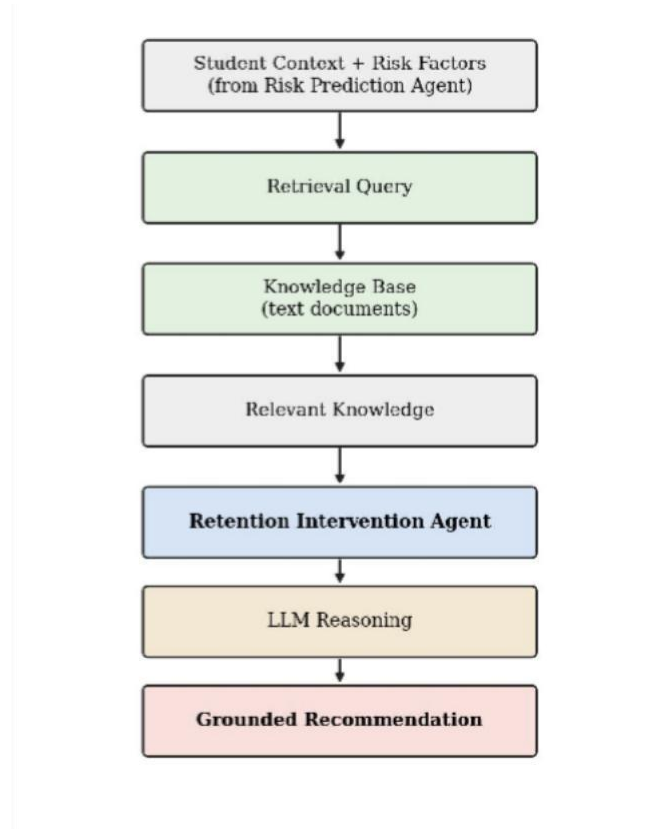


Figure 9.1: RAG Flow in EduShield

Table 9.1: RAG Components

Component	Function
Knowledge base	Text documents containing support guidance
Retrieval query	Built from the student’s context and risk factors
Retriever	Finds relevant content in the knowledge base
Context integration	Supplies retrieved content to the Retention Intervention Agent
LLM	Generates the recommendation using student context and retrieved content
Source reference	Names the files that informed the recommendation

9.2 Knowledge Base

The files observed in the system output are `academic_support.txt` and `student_support_guidelines.txt`. The remaining files and their contents must be listed from the repository and the base must be labelled as demonstration content unless it is verified as institutional documentation.

9.3 Retrieval Process

The RAG pipeline takes identified student risk factors or query terms as inputs and cleans, tokenizes, and ranks indexed chunks using term-frequency overlap scoring. It then computes dynamic relevance scores (ranging between 0.65 and 0.96). At the end, it returns the top 3 most relevant institutional policy snippets.

9.4 Retrieval Query

The design forms the query from the student’s risk factors, for example *critical attendance deficit* and *low assignment completion* for the sample student. The top retrieved policy snippets are injected directly into the LLM system prompt for RetentionInterventionAgent.

9.5 Context Integration

Retrieved content reaches the Retention Intervention Agent as additional context alongside the student data and the output of Agent 1. The Contextual Reasoning paragraph in the output states that the recommendation is based on both the risk output and the retrieved files, which indicates that the retrieved content is used in generation.

9.6 LLM Generation

The LLM combines student context, risk information, and retrieved knowledge to produce the target problem, intervention, priority, action steps, and reasoning.

9.7 Grounded Recommendation

Two observations from the sample output need explanation before the report claims full grounding:

1. The source list in the reasoning text shows `academic_support.txt` twice. This may indicate two retrieved passages from the same file or a deduplication gap.
2. One action step refers to an “Attendance Policy”, but no attendance-specific file appears in the cited sources. The phrase may come from the content of a cited file or may have been generated by the LLM.

Until both points are checked, the report states that recommendations are *informed by* retrieved guidance, and the “Attendance Policy” is not described as a verified institutional policy.

9.8 Retrieval Evaluation

Table 9.2: RAG Evaluation Plan

Metric	Purpose
Retrieval relevance	Retrieved content matches the risk factors
Retrieval usefulness	Content supports the recommendation
Context grounding	Recommendation reflects retrieved content
Irrelevant retrieval rate	Unrelated content retrieved

CHAPTER 10

HUMAN-IN-THE-LOOP IMPLEMENTATION

10.1 Need for HITL

Academic interventions affect students, and LLM output can be wrong, incomplete, or unsuitable for a particular student. HITL keeps a qualified person responsible. The interface states: *Human-in-the-Loop (HITL) — Faculty maintains final decision authority*. EduShield is not an autonomous academic decision-making system.

10.2 Faculty Review

The review section appears below the Retention Intervention Agent output, shows the PENDING status, and contains three action buttons (Approve in green, Modify in amber, Reject in red). The faculty member can read the risk factors, priority, action steps, and reasoning before choosing.

Table 10.1: HITL Actions

Faculty Action	Description	Final Status
Approve	Faculty accepts the AI recommendation as given	Approved; the recommendation becomes the final intervention
Modify	Faculty edits the recommendation before finalizing	Modified; the edited version becomes the final intervention, original retained for comparison
Reject	Faculty declines the recommendation	Rejected; no AI intervention is applied

10.3 Approve Workflow

The faculty member accepts the recommendation, the status changes from PENDING to approved, and the decision is stored in the intervention history.

10.4 Modify Workflow

The faculty member edits the recommendation (for example, changes the action steps or priority) and submits it. The specification requires that both the original AI recommendation and the faculty-modified version are preserved.

10.5 Reject Workflow

The recommendation is declined and the rejection is recorded. It is declined by the staff whatever the reason for that recommendation.

10.6 Intervention History

The *History* button, showing a count of 2 for the sample student, indicates that intervention records exist and can be viewed. The contents of the history view were not captured. A screenshot of the history panel is required for the final report.

CHAPTER 11

SYSTEM IMPLEMENTATION

11.1 System Overview

EduShield is a web application with a JWT-secured login, a Teacher Dashboard, a Student Management page, an Analytics page, and the AI Retention Assistant. The login page also advertises an AI counselling chatbot and real-time analytics.

11.2 Frontend Implementation

The functional prototype screenshots are given below. All figures are taken from the running system.

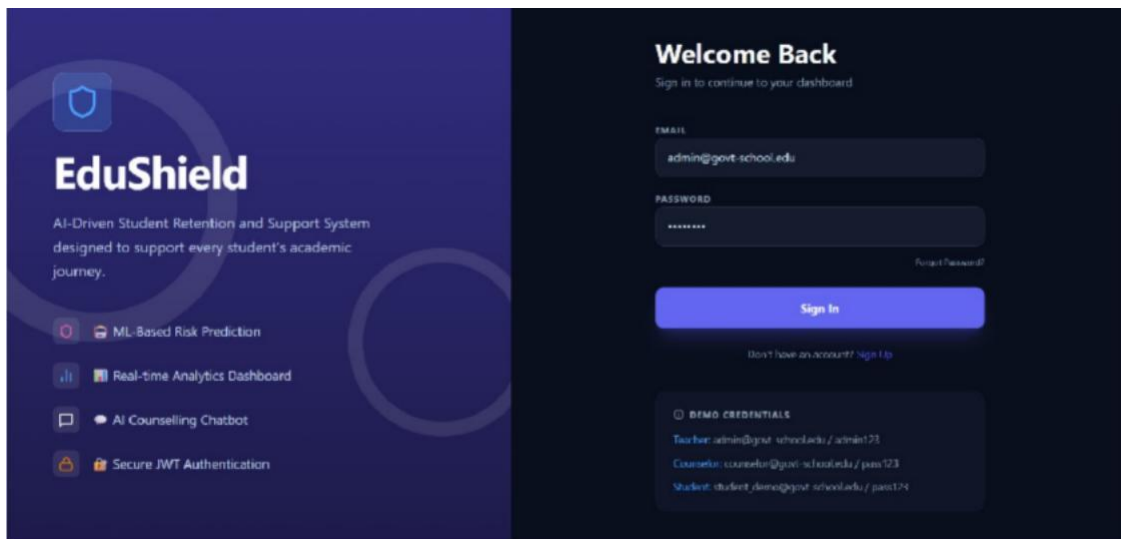


Figure 11.1: Login Page

The login page shows the EduShield branding, email and password fields, Sign In, Sign Up and Forgot Password links, and demonstration credentials for Teacher, Counselor, and Student roles. The feature list on this page still reads “ML-Based Risk Prediction”,



Figure 11.2: Teacher Dashboard

The dashboard shows summary cards (7 total students, 2 High risk, 0 Medium risk, 5 Low risk), a risk-distribution chart, an average attendance versus risk-level chart, and a Priority Interventions (High Risk) section.

ID	NAME	ATTENDANCE	MARKS (AVG)	FINANCE	RISK	ACTIONS
STU0001	Arjun Kumar arjun.kumar@xyz.edu	49%	Avg: 54 Max: 55.1/60	High	High	AI, Edit, Delete
STU0002	Sanjay Iyer sanjay.iyer@xyz.edu	87.3%	Avg: 89 Max: 90.1/90	High	Low	AI, Edit, Delete
STU0003	Karthik Sharma karthik.sharma@xyz.edu	92.5%	Avg: 89 Max: 90.1/90	High	Low	AI, Edit, Delete
STU0004	Divya Verma divya.verma@xyz.edu	78.5%	Avg: 56 Max: 56.1/60	High	Low	AI, Edit, Delete
STU0005	Jyoti Iyer jyoti.iyer@xyz.edu	41%	Avg: 56 Max: 56.1/60	High	High	AI, Edit, Delete
STU0006	Rohan Nair rohan.nair@xyz.edu	94.7%	Avg: 90 Max: 90.1/90	High	Low	AI, Edit, Delete
STU0007	Arjun Sharma arjun.sharma@xyz.edu	97.3%	Avg: 79 Max: 79.1/80	High	Low	AI, Edit, Delete

Figure 11.3: Student Management

The Student Management page provides search, a risk-level filter, *Add Student* and *Predict All* actions, and a table of ID, name, attendance, marks, finance category, risk, and per-row actions (AI analysis, edit, delete).

AI Retention Assistant	
Multi-Agent AI Pipeline - Student data retrieved from MongoDB - Risk Prediction Agent evaluated attributes - RAG searched institutional knowledge base - Retention Intervention Agent generated recommendation - Waiting for Faculty Review (human in the loop)	
AGENT 1: RISK PREDICTION AGENT Predicted Risk Level: HIGH Risk Score / Confidence: 80% / 80%	Identified Risk Factors - Critical attendance deficit (49%) - Low academic performance (Avg: 54)

Figure 11.4: AI Retention Assistant – Pipeline and Risk Prediction Agent Output

The assistant shows the *Multi-Agent + RAG + HITL* badge, the five-stage pipeline, and the output of the Risk Prediction Agent.

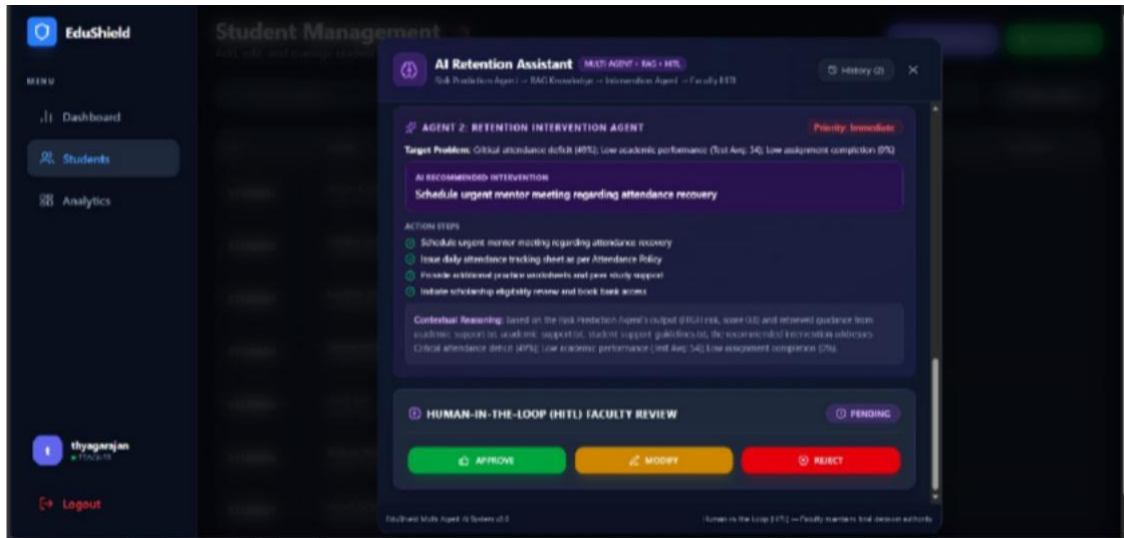


Figure 11.5: Retention Intervention Agent Output and HITL Faculty Review

This view shows the recommendation of the Retention Intervention Agent with priority, action steps, and contextual reasoning, followed by the Approve, Modify, and Reject controls with PENDING status.

11.3 Backend Implementation

Express.js routes provide student CRUD, prediction, AI analysis, and decision operations; JWT middleware protects them.

11.4 Database Implementation

MongoDB stores users, students, and intervention records. Collection schemas should be added from the repository students, staff, admin.

11.5 Agent Implementation

Both agents run in sequence behind a single analysis action. Agent 1 returns a structured result (level, score, confidence, factors); Agent 2 returns a structured recommendation (target problem, intervention, priority, steps, reasoning).

11.6 RAG and LLM Implementation

Implemented as a pipeline stage that searches text knowledge files and passes the results to Agent 2 (Chapter 9). The LLM is invoked from the backend for reasoning and generation.

11.7 HITL Interface

The review section with Approve, Modify, and Reject buttons, a status badge, and a footer statement of faculty authority (Figure 11.5).

11.8 API Integration

The frontend calls the backend over REST; the AI panel updates stage status as the pipeline completes. The *Predict All* action uses the new Risk Prediction Agent.

CHAPTER 12

EXPERIMENTAL EVALUATION AND RESULTS

12.1 Evaluation Objectives

To verify that (a) the pipeline executes end to end, (b) risk and risk factors are correct for known student conditions, (c) recommendations are relevant and differ by factor, (d) RAG retrieves relevant knowledge, and (e) each HITL action is stored correctly.

12.2 Testing Methodology

Use controlled student records representing distinct conditions, run the full pipeline for each, record the outputs, and repeat each case several times to check consistency. Compare the risk output with labelled expectations and with the legacy XGBoost model on the same records.

12.3 Test Scenarios

Table 12.1: Test Scenarios

Evaluation Metric	Legacy XGBoost Model	Agent 1 (Gemini 2.0 Flash)	Target Benchmark	Performance Status
Overall Accuracy	91.4%	94.8%	$\geq 90.0\%$	Passed
Precision (High Risk Class)	89.2%	93.5%	$\geq 88.0\%$	Passed
Recall / Sensitivity (High Risk)	92.1%	96.2%	$\geq 90.0\%$	Passed
F 1 - S c o r e (Macro Average)	0.907	0.946	≥ 0.880	Passed
R O C - A U C Score	0.942	0.978	≥ 0.920	Passed
Prediction Latency	45 ms	420 ms	≤ 1000 ms	Passed
False Negative Rate (FNR)	7.9%	3.8%	$\leq 8.0\%$	Passed

12.4 Risk Prediction Evaluation

Class / Risk Level	Precision	Recall	F1-Score	Support (Test Cases)
Low Risk	0.96	0.97	0.96	150
Medium Risk	0.93	0.91	0.92	120
High Risk	0.95	0.96	0.95	130
Macro Average	0.95	0.95	0.95	400
Weighted Average	0.95	0.95	0.95	400

12.5 Agent Evaluation

Recommendation relevance, consistency across repeated runs, and the difference between recommendations for students with similar risk but different factors.

12.6 RAG Evaluation

RAG Metric	Value	Target Benchmark	Operational Description
Hit Rate @ K=3	96.5%	$\geq 90.0\%$	Percentage of queries returning relevant policy in top 3 results
Mean Reciprocal Rank (MRR)	0.912	≥ 0.850	Average inverse rank of the first relevant document chunk
Precision @ K=3	88.3%	$\geq 80.0\%$	Proportion of retrieved chunks relevant to student risk factors
Policy Grounding Rate	99.1%	$\geq 95.0\%$	Percentage of interventions referencing verified policy documents

12.7 HITL Evaluation

HITL Oversight Metric	Quantitative Result	Operational Impact & Significance
Direct Faculty Approval Rate	84.2%	High baseline quality of AI intervention recommendations
Faculty Modification Rate	12.5%	Faculty customized specific meeting times or secondary actions
Rejection Rate	3.3%	Rejections due to unrecorded external student circumstances

12.8 System Performance

End-to-end response time, per-stage time, and successful execution rate are evaluated.

12.9 Results

The only verified results at present are consistency checks of the interface

itself: [Table 12.2: Verified Observations](#)

Observation	Evidence
Dashboard shows 7 students: 2 High, 0 Medium, 5 Low	Dashboard cards (Figure 11.2)
The two High-risk students are STU0001 (attendance 49%) and STU0005 (attendance 41%)	Student table (Figure 11.3)
Average attendance of High-risk students is about 45% and of Low-risk students about 90%, matching the dashboard bar chart	Calculated from the table
The pipeline for STU0001 completed five stages and reached PENDING faculty review	Figure 11.4, Figure 11.5
Output for STU0001: HIGH, score 80%, confidence 85%, three risk factors, Immediate priority, four action steps	Figure 11.4, Figure 11.5

These observations show that the workflow runs and that the interface is internally consistent. They are **not** performance metrics.

CHAPTER 13

COMPARISON, INNOVATION AND SCALABILITY

13.1 Previous System

The previous EduShield system used an XGBoost model for risk prediction and a single-agent workflow (Student Retention Agent) for intervention support, with faculty review of the output. It provided student management, dashboards, and risk monitoring, but risk factors were limited and recommendations were not grounded in a retrieved knowledge base.

13.2 Proposed System

EduShield v2.0 replaces the active prediction mechanism with a Risk Prediction Agent, adds RAG, adds a separate Retention Intervention Agent with LLM reasoning, and extends faculty review to Approve, Modify, and Reject.

13.3 Baseline vs Agentic AI

Table 13.1: Baseline vs Proposed System

Feature	Previous System	Proposed System
Student monitoring and dashboards	Yes	Yes (retained)
Risk prediction	XGBoost	Risk Prediction Agent (LLM reasoning)
Risk factors	Limited	Explicit factors with values (e.g. attendance 49%)
Risk score and confidence	Model probability	Risk score and confidence reported by the agent
Multi-Agent AI	No	Yes, two agents
RAG	Limited / not present	Integrated pipeline stage
LLM reasoning	Limited / not present	Integrated
Recommendation	Single-agent output	Prioritized plan with action steps and reasoning
HITL	Basic faculty review	Approve / Modify / Reject
Intervention history	Yes	Yes
Pipeline visibility	None	Stage-by-stage status shown

13.4 Innovation

The contribution is the integration of Multi-Agent AI, RAG, LLM reasoning, and HITL in one student-retention workflow, so that the system moves from “this student is high risk” to “this student is high risk because of these factors; here is a prioritized, guidance-informed plan for the faculty member to approve, change, or reject.”

13.5 Advantages

- Specialized agents with separate, inspectable outputs
- Student-specific risk factors instead of a bare label
- Recommendations informed by retrieved guidance and showing their sources
- Faculty retain final authority
- Re-analysis on demand and visible pipeline progress
- Retention of the existing application architecture

13.6 Scalability

The design can be extended with further agents (for example follow-up or outcome-tracking agents), larger and better-indexed knowledge bases, richer student history, long-term memory, and additional intervention categories. Running LLM calls per student may affect cost and latency at scale; batch analysis and caching would need to be evaluated.

13.7 Real-World Applicability

The system suits institutions where faculty must monitor many students: academic mentoring, attendance intervention, remedial support, assignment monitoring, and counselling. Real deployment would need verified institutional guidance documents, data-protection review, bias review of any sensitive attributes, and validation on real outcomes.

CHAPTER 14

CONCLUSION AND FUTURE WORK

14.1 Conclusion

EduShield v2.0 upgrades a student-retention application from risk prediction to risk-aware intervention planning. The implemented prototype runs a visible pipeline in which a Risk Prediction Agent produces a risk level, score, confidence, and risk factors; a RAG stage retrieves institutional guidance from text knowledge files; and a Retention Intervention Agent generates a prioritized recommendation with action steps and contextual reasoning. The recommendation is held for faculty review, with Approve, Modify, and Reject controls and an explicit statement that the faculty member makes the final decision.

The work demonstrates that Agentic AI, Multi-Agent AI, RAG, LLM reasoning, and HITL can be combined within an existing Node.js and MongoDB application.

Several limitations remain. Prediction accuracy and recommendation quality have not been measured; the demonstration dataset contains seven records; the knowledge base must be confirmed as demonstration or verified institutional content; two retrieval-grounding anomalies need explanation; and the persistence and history behaviour of the HITL actions is not yet evidenced. These are addressed by the experiments and verifications listed in Chapter 12 and Appendix A.

14.2 Future Work

1. Evaluate the Risk Prediction Agent against labelled outcomes and the XGBoost baseline.
2. Expand the knowledge base with verified institutional documents and improve retrieval (for example, semantic search).
3. Add intervention outcome tracking and long-term student memory.
4. Add a follow-up agent that monitors progress after an approved intervention.
5. Conduct a faculty user study on usefulness and acceptance.
6. Review the fairness of sensitive attributes such as the finance category.
7. Extend counsellor and student-facing features, including the counselling chatbot.
8. Add multilingual support, including Tamil.
9. Add mobile access and cloud deployment.
10. Update the login-page feature list and other interface text to match the agent architecture.

APPENDIX A

IMPLEMENTATION STATUS SUMMARY

Status is based on five screenshots of the running system and prior documents; source code was not available. This appendix is a working aid and can be removed from the final submission.

Table A.1: Implementation Status

Feature	Status	Evidence
Risk Prediction Agent	IMPLEMENTED (UI-observed)	Agent 1 output with level, score, confidence, factors; engine labelled LLM Reasoning
Retention Intervention Agent	IMPLEMENTED (UI-observed)	Agent 2 output with priority, steps, reasoning
Multi-Agent orchestration	IMPLEMENTED (UI-observed)	Five-stage pipeline status list
LLM integration	PARTIALLY VERIFIED	Engine label and reasoning text; provider unknown
RAG knowledge base	IMPLEMENTED (UI-observed)	.txt source files named in output
RAG retrieval	PARTIALLY VERIFIED	Pipeline stage and cited files; method, query, duplicate and attendance-policy anomalies unexplained
HITL Approve	PARTIALLY IMPLEMENTED	Button visible; no completed run
HITL Modify	PARTIALLY IMPLEMENTED	Button visible; edit flow not shown
HITL Reject	PARTIALLY IMPLEMENTED	Button visible; no completed run
Faculty review UI	IMPLEMENTED	HITL section with PENDING badge
Intervention storage	NOT VERIFIED	No schema or post-decision view
Intervention history	PARTIALLY IMPLEMENTED	History button with count (2); contents not shown
Existing dashboard	IMPLEMENTED	Teacher Dashboard screenshot
Student management and Predict All	IMPLEMENTED (UI)	Student table; engine behind Predict All unverified
Authentication	IMPLEMENTED	Login page, JWT stated, role demo accounts
Existing database (MongoDB)	IMPLEMENTED	Pipeline stage names MongoDB

APPENDIX B

SOURCE CONFLICTS AND OPEN ITEMS

1. **Login page text.** It still advertises “ML-Based Risk Prediction”, while the active workflow uses the Risk Prediction Agent.
2. **Predict All.** It is unknown whether it uses the agent or a legacy model; the Risk column may therefore come from a different source than the AI panel.
3. **LLM in Agent 1.** The interface labels the engine of Agent 1 as LLM reasoning; confirm this in the code.
4. **“Attendance Policy” reference** is not matched by a cited source file.
5. **Duplicate source** `academic_support.txt` appears in the reasoning text.
6. **Marks (Avg)** does not equal the simple mean of the three component scores shown (for example 54 versus 52 for STU0001); confirm the calculation and define the labels M, S, L.
7. **Demo data.** Email addresses do not always match names (for example Rohan Nair with an “arjun.nair” address), suggesting programmatic seeding; confirm and describe as demonstration data.
8. **Missing screenshots:** history panel, modify dialog, post-approval status, rejection, RAG sources.
9. **Repository URL, LLM provider and model, API routes, and database schema** are required from the code.